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Utilizing deep learning towards real-time snow cover detection and energy loss estimation for solar modules

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HIGHLIGHTS

- Developed a method to detect snow on solar panels in real-time.
- Model identifies snow with high accuracy of 81% on PV surface.
- Precisely estimates energy loss due to snow under 5% error.
- Demonstrates a 44% improvement over conventional computer vision methods.
- Study enhances PV system efficiency, management, and maintenance.

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ABSTRACT

Conversion of solar energy using photovoltaic (PV) panels faces challenges due to snow accumulation on PV surface in cold regions. Despite existing methods to assess this impact, there remains a gap in real-time detection and accurate quantification of the energy loss. This study introduces a novel deep learning-based method for detecting snow coverage on PV panels for maximizing solar energy conversion. The model achieved a Dice score of 0.81 when trained on a diverse dataset of PV images, achieving a 44% improvement over conventional computer vision methods. Energy losses from snow on solar panels showed the model's predictions align closely with ground-truth data, achieving an error under 5% in snow coverage prediction. Six PV arrays were analyzed for energy loss estimation under varying snow coverage conditions. The analysis showed that the model could reliably predict energy losses due to snow accumulation with a mean error of 0.05 kWh/m^2/month. The maximum energy loss was 0.23 kWh from a large PV array system covering 117 m^2 area. Analysis of the impact of snow coverage duration on energy loss showed a saving potential of 0.13 kWh/m^2 using timely clearing of snow coverage. This study highlights the effectiveness of CNN-based models in the early detection and measurement of snow coverage for improving the management and maintenance of PV systems.

1. Introduction

The integration of solar energy as a sustainable energy source is becoming more prominent with each passing year [1,2]. Rooftop photovoltaic (PV) systems offers a cost-effective strategy for decarbonization in urban energy infrastructures [3]. According to a recent study, the installation of PV panels will result in power generation of up to 563.77 GW in Poland alone [4]. This growing reliance on solar power highlights the need for efficient management of PV systems. One of the biggest challenges in achieving maximum power output is the significant decrease in performance, up to 54.5%, due to snow accumulation [5]. Since the prevalence of snow reduces the generation output of PV, the timely detection of snow cover on a PV is of critical importance for improving overall energy output [6,7]. The variability and unpredictability of PV power generation due to obstructions can limit its performance and pose challenges to the maintenance and management of the system [3]. In cold climates, where snow cover can last for days or weeks and result in up to a 34% energy loss, timely detection is crucial for maximizing the energy output of PV systems to minimize these losses [8,9]. The weight and moisture of the snow can cause micro-cracks and stress on the panel surfaces, potentially shortening their lifespan and reducing long-term efficiency [10]. In certain locations, it has been

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observed that covering of the PV surfaces leads to an energy output loss exceeding 80% [11,12]. The ability to autonomously detect snow on PV systems is crucial not only for maximizing energy output but also for maintaining the integrity of solar panels and ensuring a stable power supply. Several approaches have been developed by researchers to reduce energy losses from PV panels caused by objects obstructing sunlight from reaching the PV surface [13–15].

The amount of snow deposition on solar panels depends on factors such as the tilt of the panels, duration, and intensity of snowfall, ambient temperature, and possibly wind [16]. For instance, with a panel tilt of 14° and a long-term average annual snowfall of 124 cm, the annual electricity generation losses were 4.5%, and the average winter month electricity generation loss was 16.7% [7,17]. In another study, the total generation losses due to snow deposition were reported to be 34% of the total annual power [18]. Several studies investigated the effects of snowfall on the solar panel performance. The bypass diodes in solar PV systems can melt snow, preventing a 9% annual electricity generation loss due to snow cover on solar panels according to a study in cold climates [19]. Moreover, snow-reflected radiation affects PV performance, with shallow snow layers reducing energy output by 13% on clear days and up to 40% on cloudy days [17]. Studies have shown that other factors such as depth and location of snow on PV can impact the power output [20]. Partial shading [21] is prevalent and occurs when one or more cells within a PV array are covered, causing an imbalance in solar radiation exposure across the system [22]. Temporary coverage, caused by transient elements like seasonal snow, dust, fallen leaves, bird droppings, or human activities, further complicates the scenario and reduces the PV performance [23,24]. These coverage instances can have a pronounced non-linear effect on the PV system [25], often being random, unpredictable, and challenging to measure with conventional techniques [26]. Coverage on solar panels leads to a 0–9% power reduction as temperature rises from 0 to 20 degrees [27]. This shows an important need for an automated inspection and detection of snow cover on PV panels to detect and monitor snow coverage for timely maintenance.

Image processing methods have been investigated to automate snow detection in images [28]. Snow cover extent was detected using spaceborne synthetic aperture radar data and a random forest-based classification strategy, achieving an accuracy above 0.75 [29]. Snow cover was also detected using terrestrial photography and a Spectral Similarity-based automated algorithm, with the performance validated against supervised methods [30]. In another recent study, direct selection and perspective transformation methods were compared for snow loss detection from PV panels with a deviation of 0.1% to 7.35% [31]. However, most image processing methods rely on predefined parameters and features, which can limit their adaptability to varying conditions such as light changes, different snow textures, and complex backgrounds [32]. This necessitates the development of more sophisticated and flexible algorithms, particularly in environments with diverse and changing weather conditions. The integration of machine learning and deep learning [33] techniques offers a promising solution to these limitations.

Deep learning is a popular choice for solving complex image-based pattern recognition and object detection problems [34,35]. U-Net [36,37] and other deep learning models have found widespread application in various domains and have exhibited robust performance in tasks such as image segmentation [38–41], high-resolution and hyperspectral image change detection [42,43], building inspection [44], fault detection in PV [45], solar irradiance estimation [46] and the detection of shadow and snow [47–51]. A bidirectional long short-term memory (BiDLSTM) network was developed for accurate estimation of wind speed detection to facilitate energy conversion [52]. A support vector machine model was used to predict daily solar radiation for maximizing solar energy output which out performed other models [53]. An automated detection system utilizing ResNet-50 [54] and computer vision is introduced for the detection of snow on PV from drone-based images [55]. However, the study doesn't estimate the area of the PV covered in snow. A deep residual neural network (DRNN) has been used to determine the dust concentration on PV panels [56]. Deep learning techniques were also used to detect surface objects such as dust, cement, bird drops, shadow, snow, and soil to improve PV generation capacity [57]. However, they classified images into one category and didn't quantify the damage. In another study a method was developed to use convolutional neural networks (CNN) for semantic segmentation and classification from RGB images to classify physical faults in PV plants automatically with an average accuracy of 70% [58].

Early detection and quantification are vital for several reasons: 1) allowing for timely maintenance optimization and 2) helping maintenance teams schedule efficient snow removal to prevent prolonged periods of reduced energy output, particularly in regions with frequent snowfall [59]. Accurate estimation of snow coverage aids in predicting and mitigating energy losses more effectively, allowing for better planning and optimization of energy resources to ensure a stable and reliable power supply [60,61]. Regular monitoring and early intervention can prevent long-term damage to PV panels caused by the weight and moisture of accumulated snow, extending the lifespan of the panels and reducing the need for costly repairs or replacements [62]. By minimizing downtime due to snow coverage, the overall efficiency and reliability of solar energy production are enhanced, which is especially beneficial for large-scale solar farms and residential installations in cold climates [63]. Additionally, maximizing the energy output of PV systems in snowy regions supports global efforts to combat climate change by reducing carbon emissions and promoting renewable energy.

In light of the above, current techniques are unable to accurately determine the ratio of snow-covered areas on PV panels for reliable and efficient of solar energy conversion. As a result, this study develops a new model that couples image processing and deep learning techniques for pixel-wise segmentation, using the U-Net model, to assist in the early detection and quantification of snow coverage. The primary data sources for this research consisted of timelapse images capturing solar panels under varying degrees of snow coverage. In terms of limitations, the methodology accounts for the amount of snow coverage and the impact on string design losses, covering aspects such as module mismatch, diodes, DC wiring, soiling, and reflection losses; however, the location and pattern of snow coverage on the PV panels was simplified by considering the coverage-area ratio. Also, the developed framework requires a reference no-snow image for accurate loss estimation. Overall, this study aims to develop an accurate method for detecting snow coverage on PV panels using deep learning in order to facilitate early snow coverage detection and optimized maintenance. This approach enhances real-time snow coverage management, contributing to the reliability of solar energy generation in cold climate.

2. Methodology

Snow accumulation on PV panels results in excessive generation energy loss for a PV panel, especially in cold regions. This study utilized a detailed methodology to evaluate the effects of snow coverage on the efficiency of PV panels. To outline the methodology used, a flow chart was developed as shown in Fig. 1. The proposed method consisted of five key steps. The first step was the collection of images of PV panels with various backgrounds to create a diverse set of images. Then each image was manually annotated to create a segmentation mask for the creation of a segmentation dataset. This dataset was used in the training of the U-Net deep learning model. The fourth step involves evaluating the trained model on the segmentation task to determine its accuracy and reliability in real-world scenarios. Finally, the fifth step was the deployment of the trained deep-learning model in a real-world environment for the estimation of snow coverage on PV panels in images.

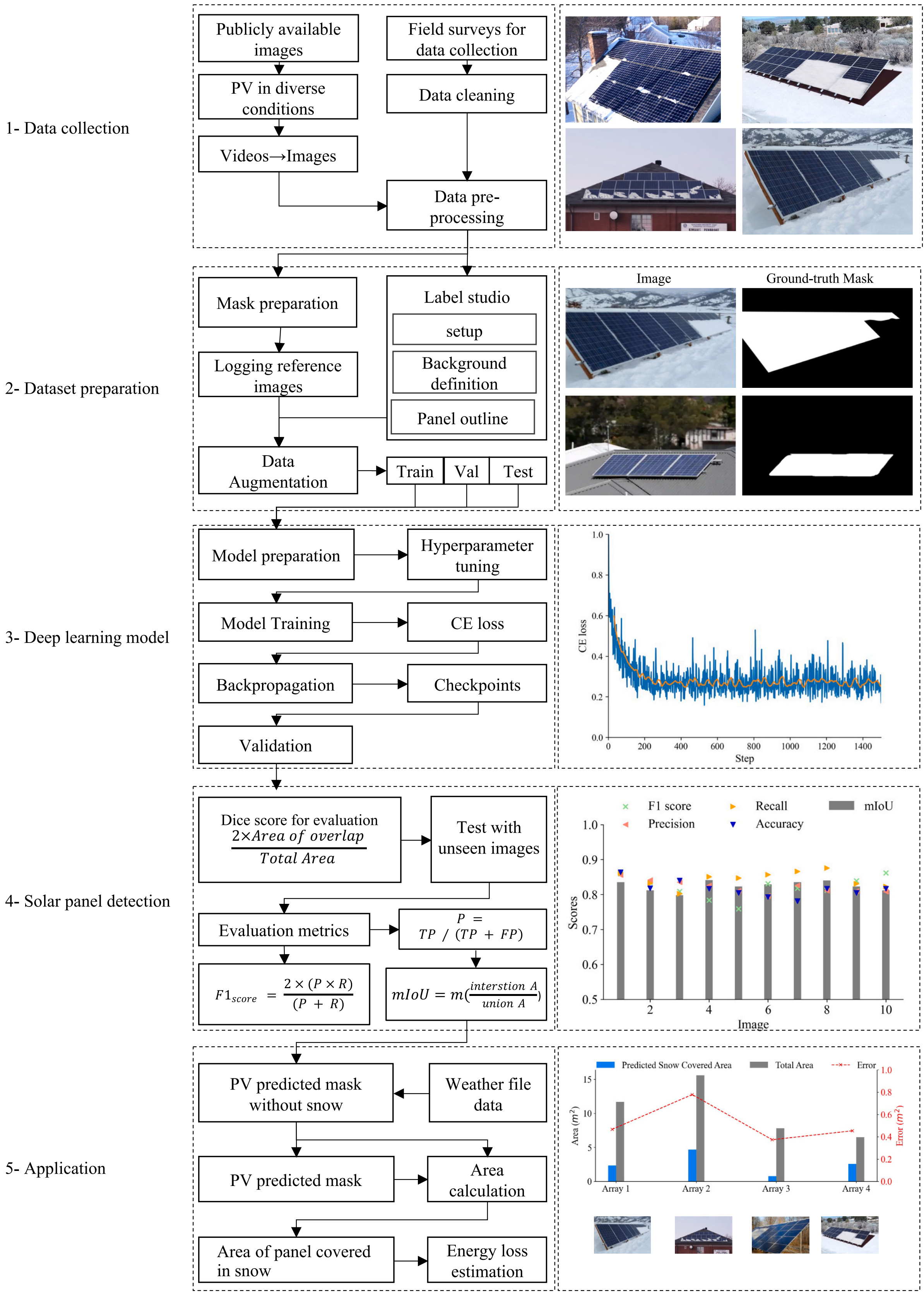

Fig. 1. Flow chart of the proposed method.

2.1. Data collection for diverse environmental conditions

The dataset was prepared from various videos and images collected from different sources as shown in Fig. 2 [64–66]. A set of 250 unique images were obtained with a wide range of diversity and environmental complexity. In addition, both small and large size images were collected. The smallest size of the image in the dataset was 512 × 512 pixels, and the largest size of the image was 3024 × 4032 pixels. Fig. 2 shows a sample of the dataset that includes images of solar panels with various backgrounds, such as roof tiles, roofs, skies with vegetation and snow coverage, walls with snow coverage, and obstructions. A set of 6 PV arrays were selected for visualization and implementation of the proposed methodology.

2.2. Preparation and augmentation of solar panel dataset

The dataset was generated from collected PV images using label studio editing software which offers image editing tools such as points selection and coloring for the dataset preparation process. The samples of the training data with ground-truth masks are shown in Fig. 3 below. The ground-truth mask is a binary black-and-white image for each PV panel image in the dataset where white pixels represent the PV panels, while black pixels represent the background, indicating areas without panels.

Two classes were considered during the manual preparation of the data, with the surface of the visible panel being labeled as one target class and the background being classified as the second class. The process of creating masks involved manually tracing the boundaries of the objects of interest within the images. Accurate and precise masks were crucial to ensure that the model could effectively learn to segment the objects of interest from the background. The ground truth mask was represented as a black-and-white image, with black pixels denoting the background and white pixels representing the PV panels in the RGB image as shown in Fig. 3.

To increase the number of image-mask pairs for the training of the model, data augmentation techniques were employed. Data augmentation refers to the diversification of the collected image data set through the application of various image transformation techniques, such as scaling, cropping, flipping, padding, rotation, translation, and affine transformation. Fig. 4 illustrates the techniques utilized for data augmentation in the case of Array 1. As a result, a set of 800 images and mask pairs were utilized during the training of the model.

2.3. Deep learning model for snow loss detection

The selection of U-Net for this task was based on its suitability as a CNN specifically designed for image segmentation tasks. In the context of segmentation, the convolution operation involves sliding a small window, known as a filter or kernel, across the image and calculating the dot product between the filter and the corresponding image pixels (as shown in Eq. (1)).

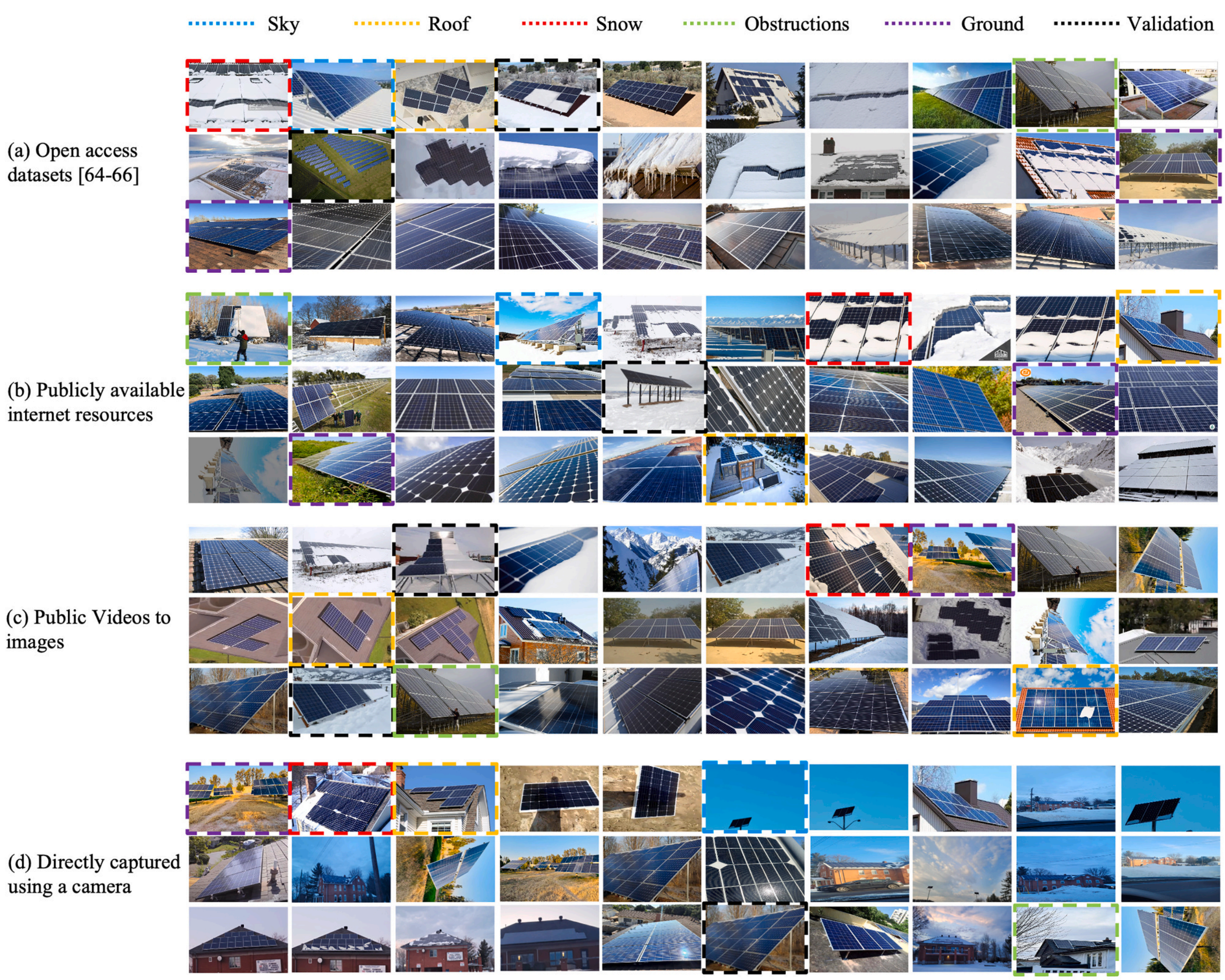

Fig. 2. Set of diverse background complexities in the developed dataset.

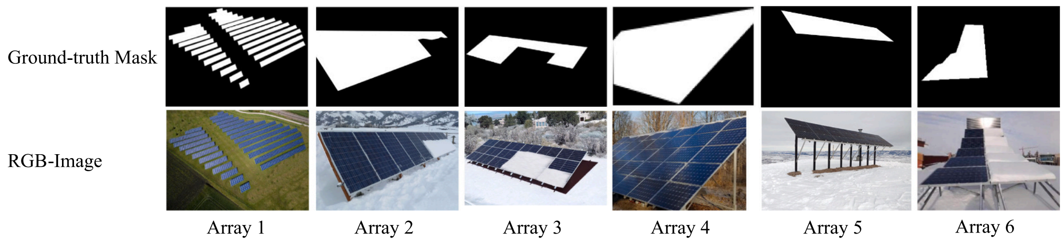

Fig. 3. Dataset sample PV arrays with ground-truth mask for each array.

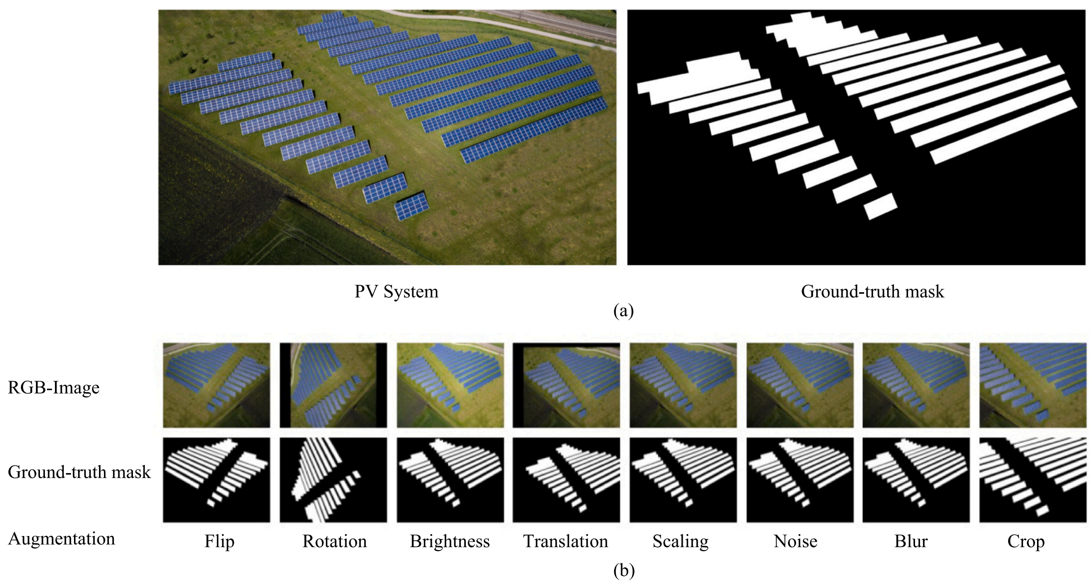

Fig. 4. Data augmentation techniques (a) original image mask pairs and (b) augmentation results.

$$i^*K = \sum\nolimits_{-k}^{k}\sum\nolimits_{-k}^{k} i(x,y) \bullet K(x+u, y+u) \tag{1}$$

Where K is the kernel or filter used in convolution operation, (x, y) shows the pixel location in the image. This dot product is then used to create a new image, known as a feature map, which contains information about the presence and strength of certain features in the input image. The output height (H_{out}) and width (W_{out}) of the convolution layer is calculated using the following Eq. (2) and Eq. (3).

$$H_{out} = \left[\frac{H_{in} + 2 \times padding - dilation \times (kernel_size - 1) - 1}{\text{stride}}\right] + 1 \tag{2}$$

$$W_{out} = \left[\frac{W_{in} + 2 \times padding - dilation \times (kernel_size - 1) - 1}{\text{stride}}\right] + 1 \tag{3}$$

Fig. 5 shows the U-Net architecture used for segmentation that was developed using Python programming language [67] and PyTorch [68] deep learning library. The implemented U-Net architecture consisted of contracting and expansive path. In the contracting path, each block consisted of two consecutive convolutions with a size of 3 × 3, followed by a ReLU activation function [69], and then a max-pooling layer. This arrangement was repeated multiple times, effectively downsizing the image from its initial dimensions of H, W, 3 to H/16, W/16, 1024. This downsizing process was used to capture and encode the contextual information within the image, allowing for a more efficient analysis of the spatial hierarchies in the data. This allowed the model to learn a hierarchical representation of the input PV image, with each layer learning increasingly complex features as the image is processed through the network.

The expansive path used bilinear interpolation to up-sample the feature map and output the desired size. Skip connections [70] were used between the up-sampling and down-sampling paths to retain important contextual information in the deep layers [70]. The skip connection provided local information to the global information during upsampling, which was concatenated with upsampling. The upsampling block of U-Net was used to increase the spatial resolution of the feature maps produced by the encoder part of the network. This allows the decoder part of the network to effectively fuse the high-level semantic information from the encoder with the low-level spatial information from the decoder, resulting in more accurate image segmentation. The final layer is the classification which uses a 1 × 1 convolution layer to reduce the number of channels to 1. The output was a mask with either 1 or 0 valued pixels representing the predicted PV or background pixels, respectively.

2.3.1. Model training

The training process aimed to enable the U-Net model to accurately detect and measure the number of PV pixels in an input RGB image. The model was trained using a dataset of 250 images with diverse backgrounds, each manually annotated. In this study, a supervised training method was adopted. An input image and its corresponding mask were provided to the model. The data underwent processing through both the contracting and expansive paths, resulting in a predicted mask. This predicted mask was then evaluated against the actual mask using a loss function to measure the accuracy of the training. Errors are then propagated back through the system, causing the system to adjust the weights which control the network. This process occurs over and over as the weights are continually tuned. During the training of a network, the same set of data is processed many times as the connection weights are refined.

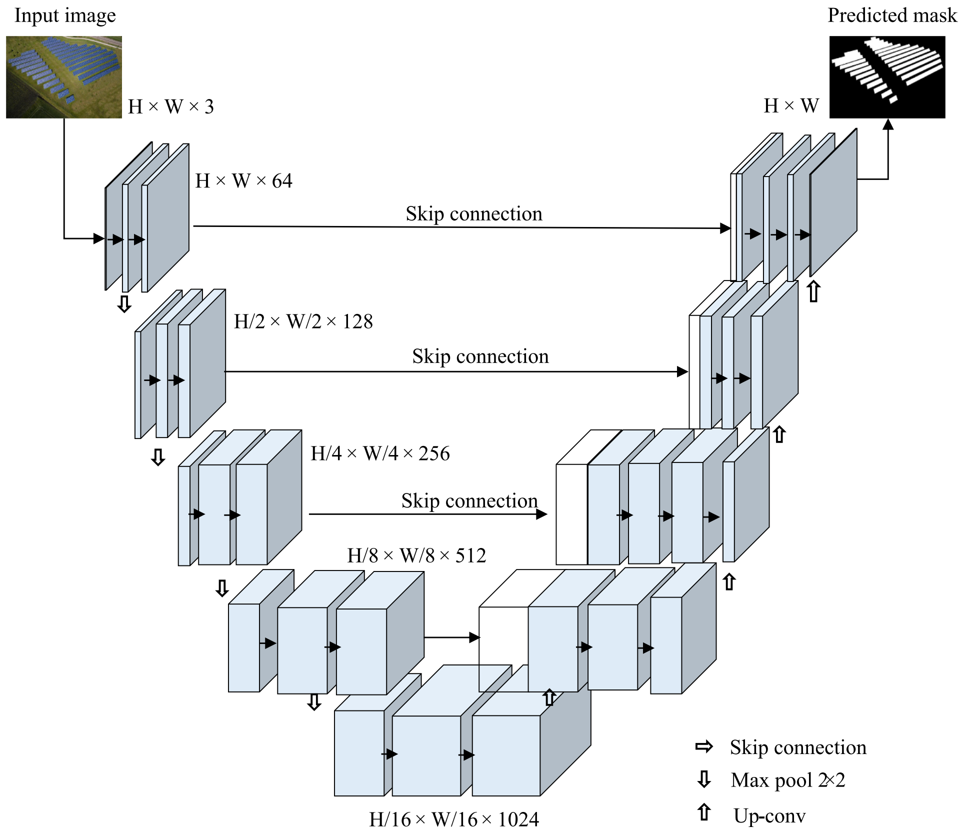

Fig. 5. U-Net architecture implementation in this study.

2.3.2. Loss function for PV detection

The loss function that was used is the binary cross entropy loss (BCE) function. The BCE is commonly used in training image segmentation models such as U-Net [71,72]. This loss function is used to measure the dissimilarity between the predicted segmentation mask and the ground-truth mask. The Eq. (4) for binary cross entropy loss is given below. The ground-truth energy loss was calculated based on the measured snow coverage, while the predicted energy loss was calculated using the snow coverage values predicted by the U-Net model.

$$\mathscr{L} = -\frac{1}{N}\sum\left[y \bullet log(p) + (1-y) \bullet log(1-p)\right] \tag{4}$$

where N is the total number of pixels in the image, y is the ground truth mask and p is the predicted mask. In the case of PV panel segmentation, y is a binary mask where white pixels correspond to PV panels and black pixels correspond to the background. p is the predicted segmentation mask from the U-Net model, also in binary format. The goal of training is to minimize this loss function so that the predicted mask is as similar as possible to the ground-truth mask. The training will be considered successful if the loss is minimized and the predictions are accurate.

2.3.3. Dynamic learning rate adjustment

Moreover, in the training of deep learning models, the learning rate is an important hyperparameter that determines the pace at which the model learns from the training data. A high learning rate can cause the model to converge quickly, but it can also lead to overshooting the optimal solution. A low learning rate can ensure that the model converges to a good solution, but it can also lead to slow convergence. One of the advantages of using a dynamic learning rate is that it can help the model converge faster and more efficiently. These approaches adaptively adjust the learning rate during training, allowing the model to efficiently navigate the landscape of the loss function. Studies have employed dynamic learning rates, showcasing their utility in achieving state-of-the-art results on complex image segmentation tasks [73]. This is because a high learning rate can be used in the early stages of training when the model is far from the optimal solution and a lower learning rate can be used in the later stages of training when the model is closer to the optimal solution. This adaptive approach can help the model avoid getting stuck in a suboptimal solution and converge to the global minimum of the loss function. Similarly, in this study, a dynamic learning rate was adopted during the training process.

2.4. Detection and validation of PV surface coverage

For model evaluation, a Dice score was used to quantify the performance on the testing dataset. The Dice score [74] is a commonly used metric for evaluating the performance of image segmentation models. The Dice score is calculated as the ratio of the intersection of the predicted and ground truth regions to the union of the predicted and ground-truth regions. The Dice score is given by Eq. (5).

$$Dice\ Coefficient = \frac{2 \times (A \cap B)}{A + B} \tag{5}$$

where A is the true mask, and B is the predicted value of the model. Moreover, Precision, F1-score, and accuracy were also used to evaluate the performance of the classification of the trained U-Net model. Precision, as indicated by Eq. (6) quantifies the proportion of true positive (TP) predictions among all positive predictions. It emphasizes the accuracy of positive predictions, providing insights into the model's ability to avoid false positives (FP).

$$Precision = \frac{TP}{(TP + FP)} \tag{6}$$

Eq. (7) shows F1-score which combines precision and recall producing a balanced metric that considers both the ability to make accurate positive predictions and to capture all positive instances. Recall is calculated using Eq. (7) where FN is false negative. Similarly, accuracy, represented by Eq. (8) assesses the overall correctness of the model's predictions. It evaluates the ratio of correct predictions (true positives and true negatives) to the total number of predictions.

$$F2 - Score = \frac{2 \times (Precision \times Recall)}{(Precision + Recall)} \tag{7}$$

$$Accuracy = \frac{(TP + TN)}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{8}$$

2.5. Energy management application

To calculate the snow-covered area on PV panels (denoted as A_{snow}), a multistep method was used. The trained U-Net model was initially used to detect PV pixels within the given input. This approach faces three main challenges in estimating the actual area of snow from an image. Firstly, the information regarding the dimensions of the PV panels is unknown, which complicates the conversion of pixel data into real-world measurements. Secondly, while deep learning models, such as U-Net, are effective in classifying pixels of PV or background, they do not inherently provide information on the scale or dimensions of the objects being classified. Thirdly, the presence of snow in the background can lead to confusion regarding whether it is on the panel or not, adding another layer of complexity to the accurate estimation of snow coverage. To address these challenges, a reference image of the PV scene before the snowfall was used Fig. 3 (b). This reference image should be from the same distance and angle as the test image and captured when no snow was present on the PV panels. Uniform camera settings across both reference and test images were essential to minimize discrepancies caused by the camera. Therefore, the developed methodology for snow area quantification is only adaptable to situations where a fixed camera is used for monitoring PV systems in the field.

2.5.1. Estimation of energy loss due to snow accumulation

Once the reference image has been stored for the testing environment, the trained model processes both the reference image and the testing image to remove the background and identify the PV pixels in the images as shown in Fig. 6. The network's task is to only determine the total number of pixels representing the PV panel in the image, providing a measure of the PV panel's area in the absence of snow (A^{pv}_{ref}). The same model processes a test image of the same PV panel, but this time with snow coverage. In this snow-covered image, the model calculates the area represented by PV panel pixels (termed A^{pv}_{test}). Due to the snow obscuring parts of the panel, A^{pv}_{test} is expected to be smaller than A^{pv}_{ref}. As shown in Fig. 6, the reference image and test image of the PV panel were given to the trained model as inputs. The model used the trained parameters to detect the PV panels and remove the background to output a black-and-white mask. Then the area of PV was calculated using Eq. (9).

$$A_{pv}(px^2) = \frac{N_{white}}{W \times H} \tag{9}$$

Where N_{white} is the total number of the white pixels in the output mask and W, H are the height and width of the image, respectively. The difference between these two values was used to calculate the area covered by snow on the PV panel (A_{snow}) as shown in Eq. (10).

$$A_{snow} = A^{pv}_{ref} - A^{pv}_{test} \tag{10}$$

2.5.2. Energy generation losses analysis

To assess the impact of snow accumulation on PV panels and its subsequent effect on energy production, a detailed mathematical model has been introduced based on [8,75]. This model quantitatively determines the relationship between snow accumulation on PV panels and the reduction in energy generation. The approach involves estimating the energy loss due to snow cover indirectly by comparing the actual energy produced by the PV system $E(H_{i,s})$, with the estimated energy production if there were no snow on the PV module $E(H_i)$ as shown in Eq. (11).

$$E_{loss} = E(H_i) - E(H_{i,s}) \tag{11}$$

where H_i is the total in-plane irradiance (kWh/m^2) and $H_{i,s}$ is the total in-plane irradiance (kWh/m^2) with snow. Based on the previous work [76], the estimated daily energy production without snow is determined using the following Eq. (12), which accounts for the effects of temperature and irradiance.

$$E_{loss} = P_o \times (H_i - H_{i,s}) \times PR_{25} \times [1 + c_p \times (T_w - 25)] \tag{12}$$

where P_o is the nameplate d.c. power (kW), PR_{25} is the system performance ratio at 25 °C PV module temperature (dimensionless), c_p is the power correction factor for temperature (°C^{-1}), with a value of −0.0045 and T_w is the irradiance-weighted PV module temperature (°C). The T_w provides a representative PV module temperature for correcting performance for temperature effects which is calculated using Eq. (13).

$$T_w = \frac{\sum T_i \bullet G_i}{\sum G_i} \tag{13}$$

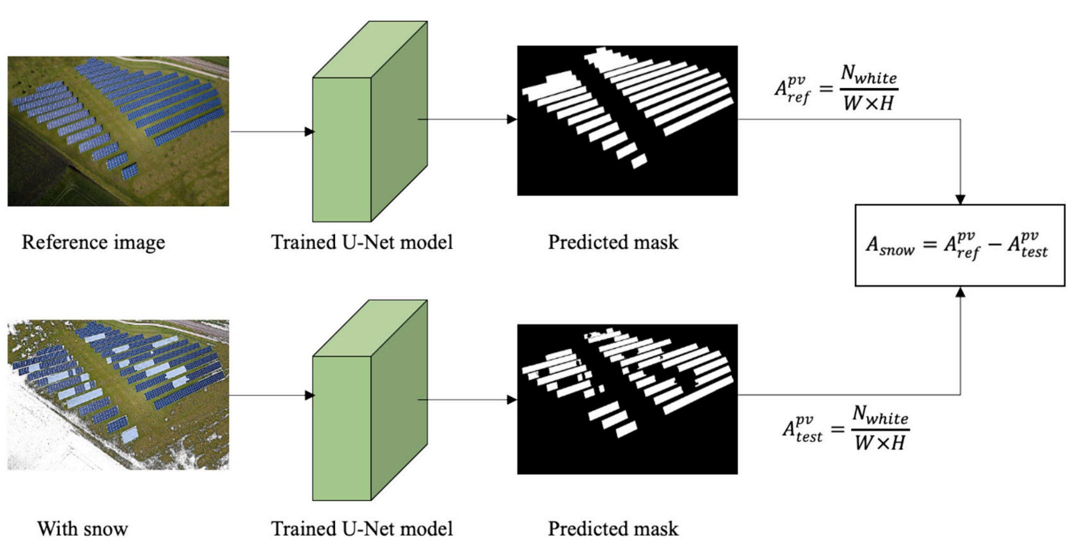

Fig. 6. Methodology for detection of snow cover percentage using trained deep learning model.

Where G_i is the plane of array (POA) irradiance (W/m^2), T_i is the PV module temperature for individual hourly measurement. The PV module cell temperature is calculated using the Sandia equations for a glass/cell/polymer PV module installed on an open rack using Eq. (14).

$$T = G \bullet e^{-3.56-0.075 \bullet s_w} + T_a + G \bullet 3/1000 \tag{14}$$

where s_w is the wind speed in (m/s) and T_a is the ambient air temperature (°C).

2.5.3. Validation of energy losses

To validate the energy losses predicted by the developed method, ground-truth energy losses and predicted energy losses were compared. The ground-truth energy losses are calculated using the actual snow coverage (A_{snow}) which was calculated manually. The predicted energy losses are based on snow coverage predicted by the U-Net model (A'_{snow}). The ground-truth E_{loss} and predicted E_{loss} are given in Eq. (15) and (16), respectively.

$$\text{Ground}-\text{truth } E_{loss} = E(H_i) - E\left(1 - \frac{A_{snow}}{A_{PV}} \times H_i\right) \tag{15}$$

$$\text{Predicted } E_{loss} = E(H_i) - E\left(1 - \frac{A'_{snow}}{A_{PV}} \times H_i\right) \tag{16}$$

Where H_i was obtained from the weather data. The amount snow was used to estimate the decrease in the total in-plane irradiance on the PV panel and reducing the amount of energy output.

2.5.4. Examined location and weather condition

For analysis of energy output from solar panels, the weather file for the city of Toronto, Canada, was utilized for the winter months (December to March) as shown in Fig. 7. The average ambient temperature varies between −14.25 °C in December to 8.2 °C in March. Average solar irradiation reaches a maximum of 4.5 kW/m^2/day in

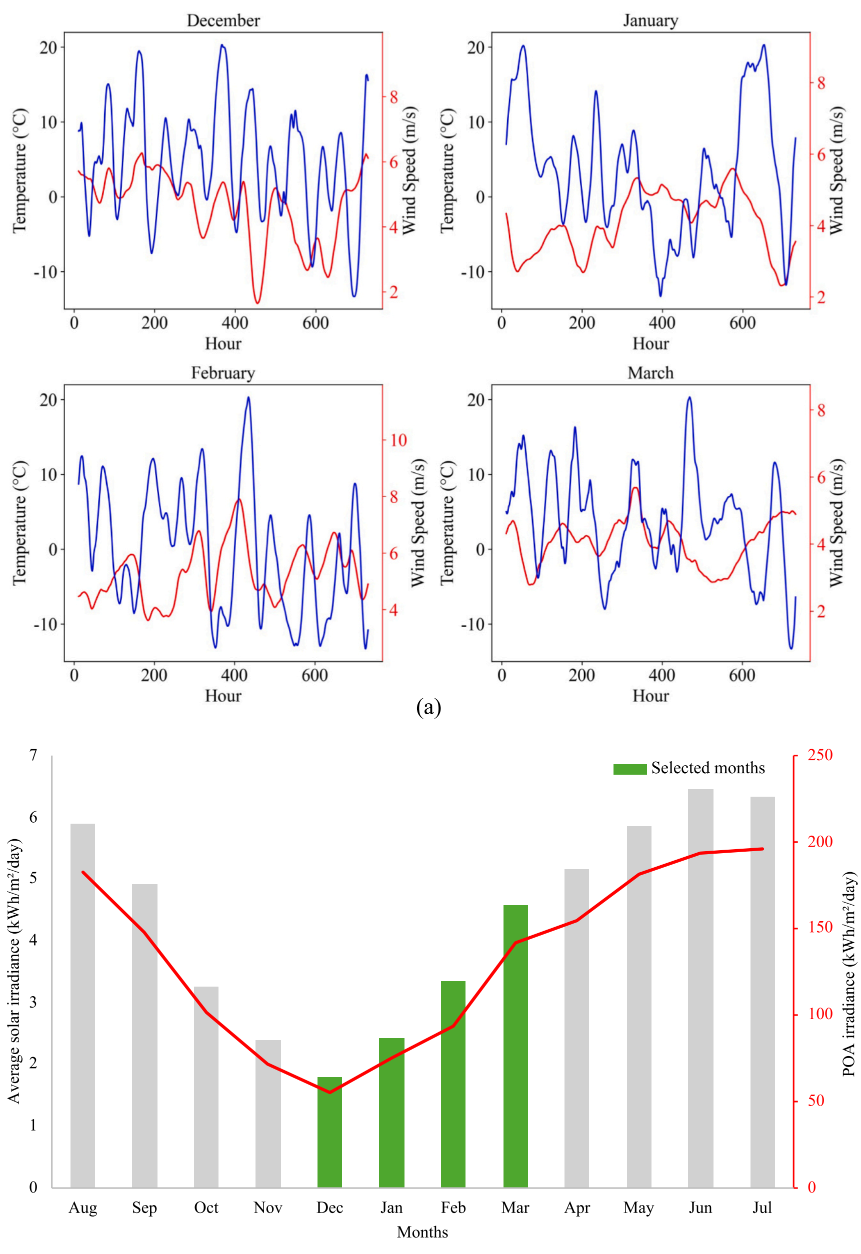

Fig. 7. Monthly data for (a) Dry bulb temperature and wind speed (b) Daily average solar irradiance and POA irradiance for Cambridge, Canada.

March while the average for winter months was 3.03 kW/m^2/day. The wind speed has an average value of 5.58 m/s throughout the winter months.

3. Results and discussion

The results and discussion section covers various aspects, including the train loss curve of the U-Net model on the PV dataset, the dynamic learning rate curve of the training process, training samples with masks, the Dice curve of the U-Net model on the PV dataset, and testing results compared to the expected ground-truth. Additionally, the section discusses the percentage of area covered by snow as an important outcome of the study.

3.1. Deep learning model training results

To assess the performance of the trained model, Dice score and loss values were monitored as shown in Fig. 8. The graph of Dice score vs epochs shows that a maximum of 0.81 was achieved during the training process. This high Dice score indicates that the model was able to accurately segment PV panels in the RGB images, with a high degree of overlap between the predicted and ground truth regions. The Dice Min and Dice Max lines on the graph represent the range of the model's performance on the validation set at each epoch, with the former indicating the lowest Dice score for the poorest-performing sample and the latter showing the highest score for the best-performing sample. The U-Net model was trained using a training dataset. The entire project was coded in Python and PyTorch library, a widely used open-source tool in the field of deep learning [68]. Computational work was carried out on two high-performance computing platforms: Google Colab and Kaggle Kernel. Each platform offered different GPU configurations to optimize processing efficiency. On Google Colab Nvidia K80 and T4 GPUs, equipped with 12GB and 16GB of GPU memory respectively, and capable of delivering 4.1 TFLOPS and 8.1 TFLOPS in performance were used. Meanwhile, Kaggle Kernel was provided with an Nvidia P100 GPU, featuring 16GB of GPU memory, a memory clock speed of 1.32GHz, and a computational performance of 9.3 TFLOPS.

3.1.1. Validation scores of the trained U-net model

Fig. 9. presents a numerical value achieved such as the precision, F1 score, and accuracy of the trained model used for PV detection in RGB images. Fig. 9 (a) illustrates the validation loss over epochs for different learning rates, showing the impact of learning rate adjustment strategies. As the learning rate increases the validation loss saw a stronger decline. In Fig. 9 (b) the results of mIoU, F1 score, precision, recall, and accuracy over the dataset, provide insights into the model's overall efficacy. The chart displays the performance metrics for 10 unseen test image samples. The highest *mIoU* achieved was fourth test image with 0.831. Fig. 9 (c), shows the precision and recall metrics derived directly from training results, highlighting their progression throughout epochs. Fig. 9 (d), shows a box plot comparison of validation loss at different learning rates and offers a representation of their performance variations. The metrics validated the successful training process of the deep learning models.

A sample of PV panels has been shown for validation of the performance of the trained model on unseen PV images. Fig. 10 (b), shows a qualitative analysis of the performance of the trained U-Net model on a sample of 6 images from the dataset. The predicted mask closely aligns with the hand-annotated ground-truth masks. Fig. 10 also quantifies the performance of the model on the same images using the Dice score. The model achieved the highest score on Array 1 with a Dice score of 0.90. A mean deviation of 0.15 was noticed for the sampled images from the true masks.

3.1.2. Comparison with traditional methods

For comparison with traditional methods, Fig. 11 (a) illustrates the testing image input alongside the output of the U-Net model. In Fig. 11, the testing image shows a simple snow background, with predicted and ground truth masks. The U-Net model adeptly removes the background and precisely detects the exposed area of the solar panels, as demonstrated. The same validation images were processed by popular image processing-based segmentation methods such as Canny edge detection [77], K-Means-based segmentation [78], and image thresholding. The comparative analysis of segmentation techniques on the dataset reveals that the U-Net model outperforms the other methods. Irrelevant features, highlighted in red rectangles, predominantly appear in methods based on image processing. In contrast, the deep learning method learns to differentiate between relevant and irrelevant details, effectively reducing noise and enhancing the accuracy of segmentation.

These results have been quantified using the Dice and mean intersection over union (mIoU) metric as shown in Fig. 11 (b) The Canny edge detection method achieved a mean Dice score of 0.60 suggesting moderate effectiveness in capturing the essential features of the objects. The K-means clustering method achieved an average Dice score of 0.47. The thresholding method demonstrates the most effectiveness with a Dice score of 0.63. The U-net outperformed all the methods with a mean Dice score of 0.91. These results show the superiority of deep learning approaches like U-Net for complex image segmentation tasks compared to unsupervised image processing techniques.

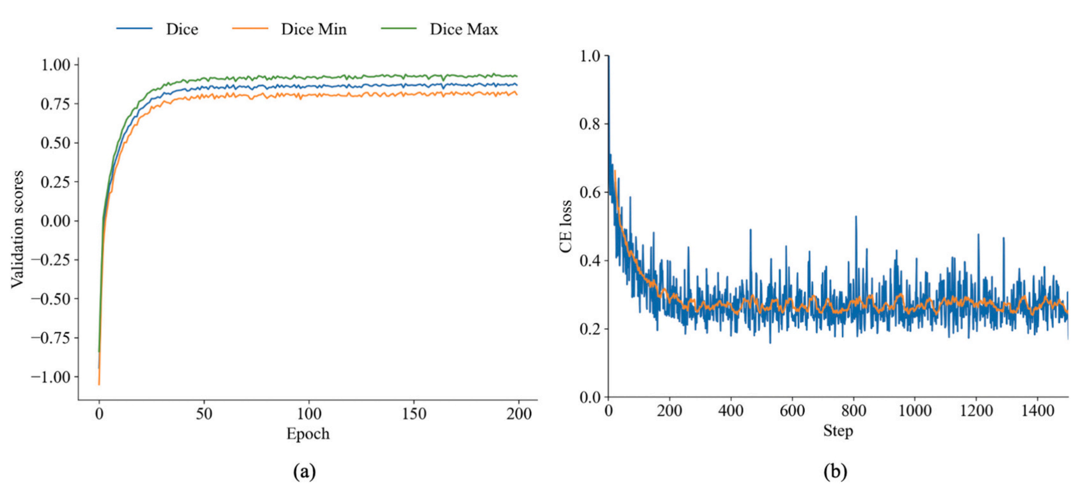

Fig. 8. (a) Validation Dice score and (b) Training CE loss of the U-Net.

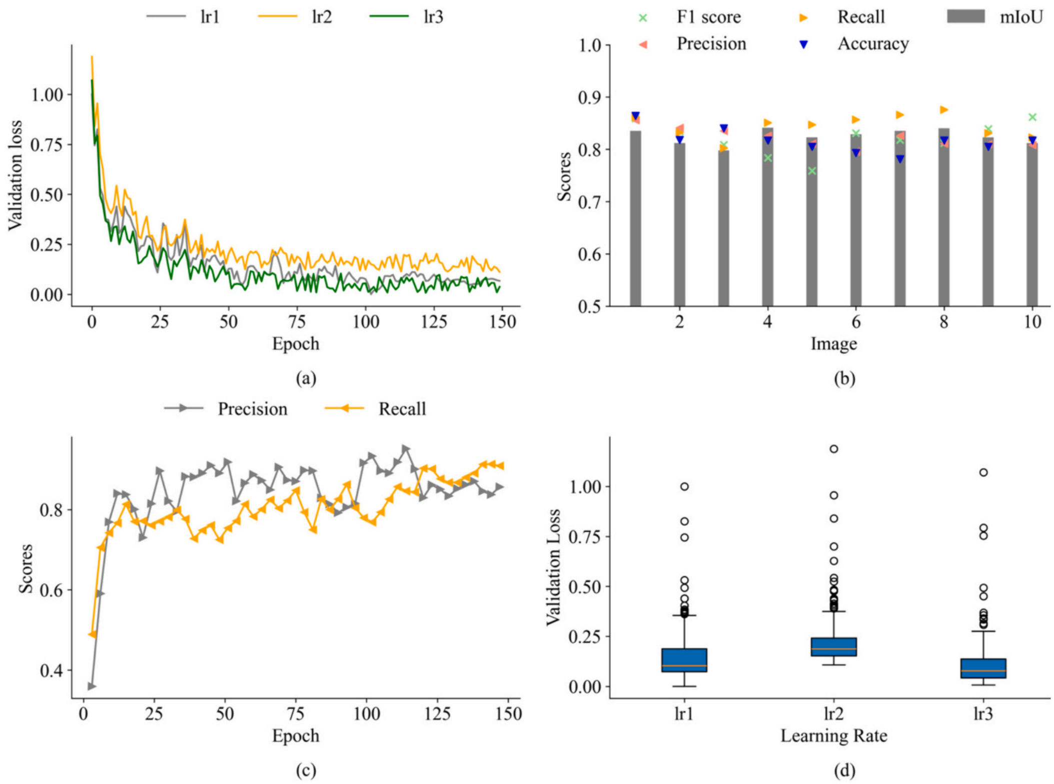

Fig. 9. Performance metrics of trained mode. (a) validation losses (b) metrics of 10 samples (c) Precision and Recall vs Epoch and (d) Validation loss distribution over different learning rates.

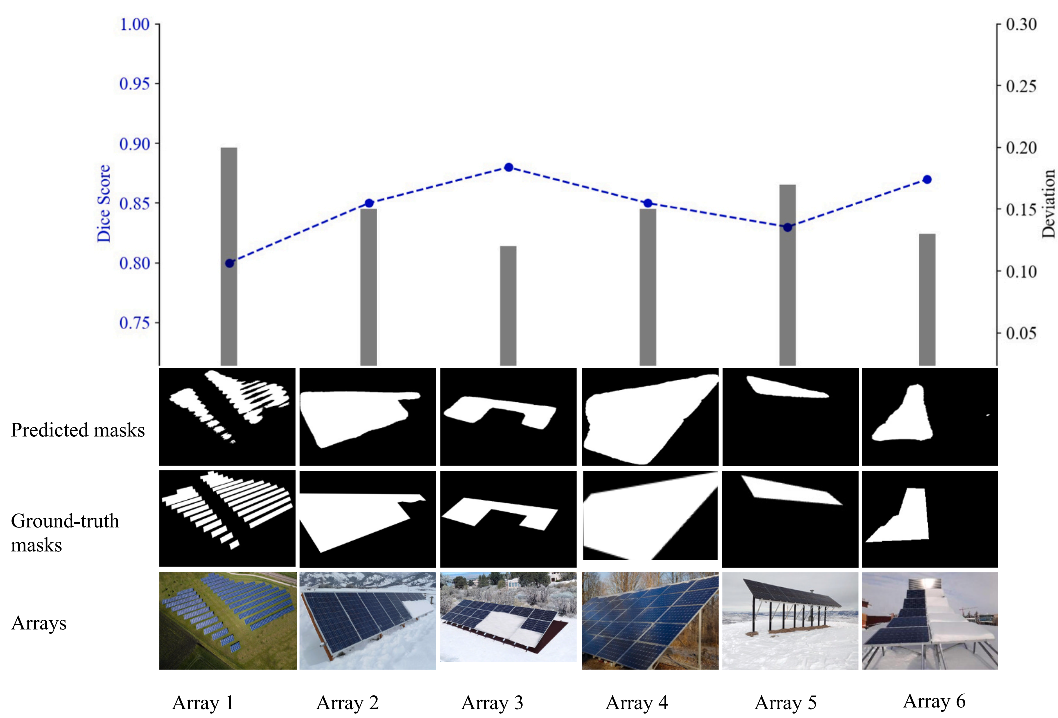

Fig. 10. Dice score achieved by trained U-Net model.

3.2. Quantitative estimation of snow coverage area

Fig. 12 (a) and (b) show the input RGB image at different times of the year which are before snow cover on PV and after snow cover on PV, respectively. Array 1, consisted of multiple panels spread over a large area, the error was larger (20 m^2 difference between actual and predicted snow cover) compared to others as limited images of such systems were present in the collected dataset. The Array 1 area is shown to be 1/

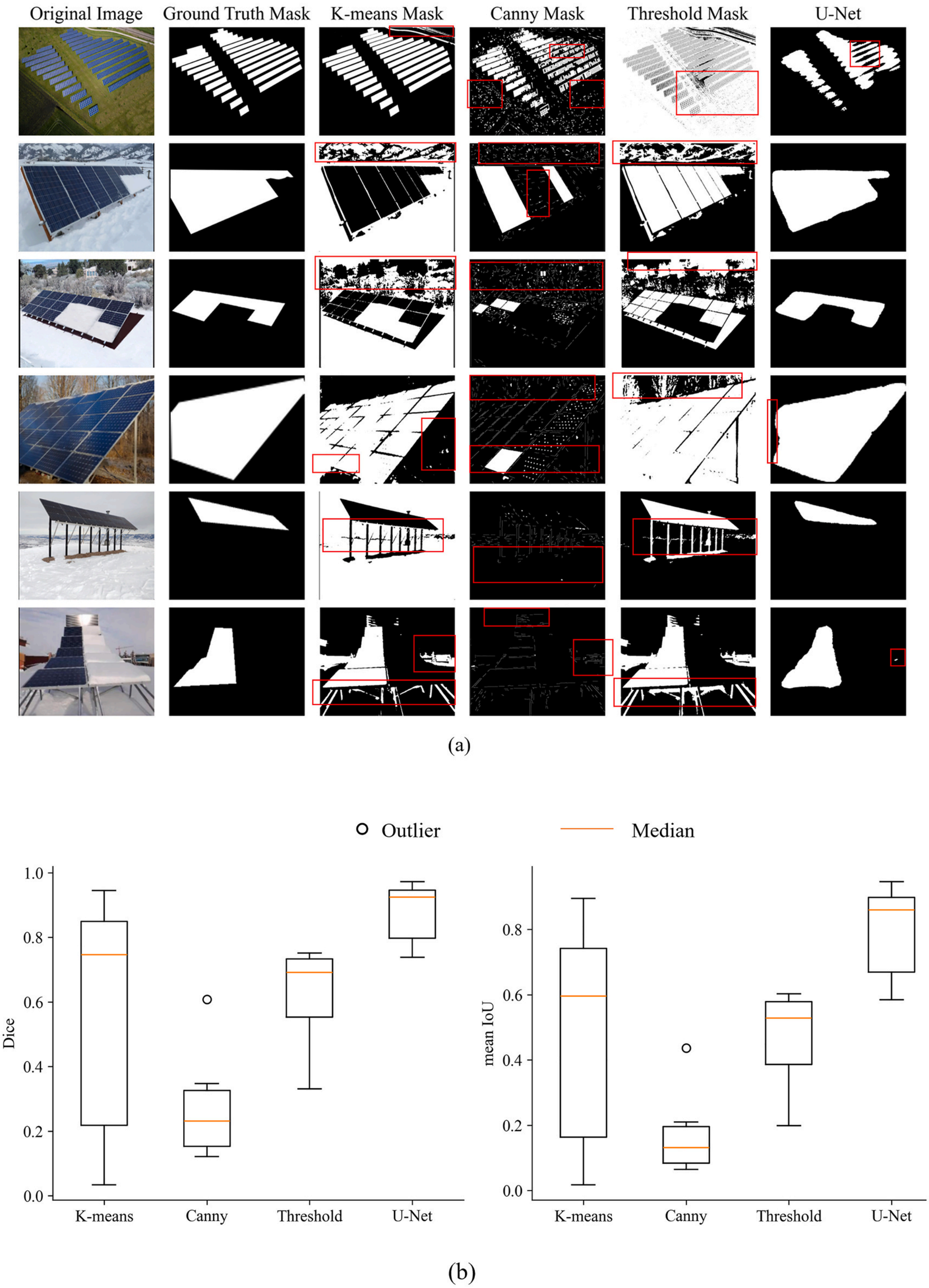

Fig. 11. Comparison with image processing methods using (a) qualitative analysis and (b) Quantitative analysis.

10 of the actual area for visualization purpose. For Array 2, situated in a cold, mountainous environment, the error significantly increased to about 0.78 m^2 due to the larger size of the array and angle of view. In settings, such as those with trees, a slightly lower error of 0.374 m^2 was observed. In the fourth array, the total error in the snow-covered area was 0.45 m^2. The prediction errors were lower when arrays were placed closer to the camera. These discrepancies underscore the complexity of accurately predicting snow coverage on PV panels, influenced by various factors including the panel's location, the surrounding environment, and seasonal changes.

3.3. Energy loss modeling in the examined arrays

The six arrays have been analyzed for energy loss analysis using four different snow coverage conditions. Fig. 13 presents the energy losses for each array, comparing the energy loss values calculated from the

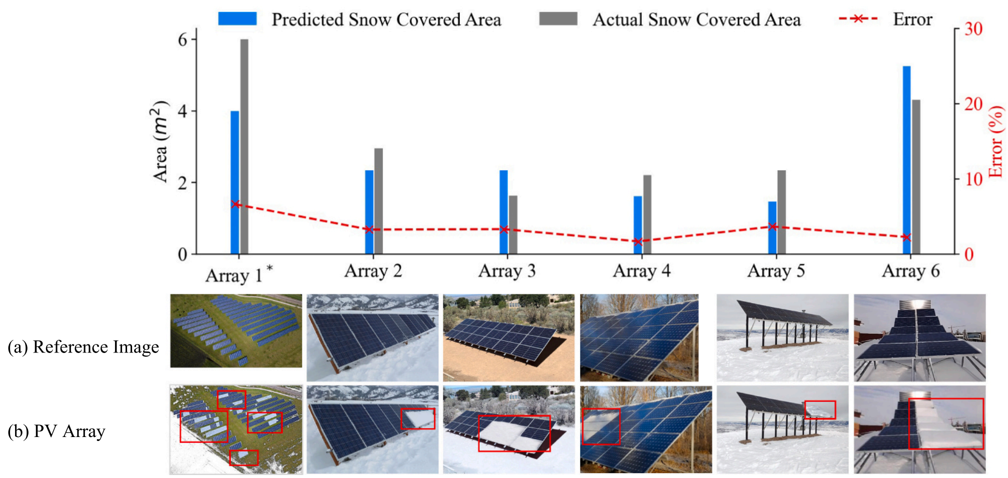

Fig. 12. Predicted area covered by snow.

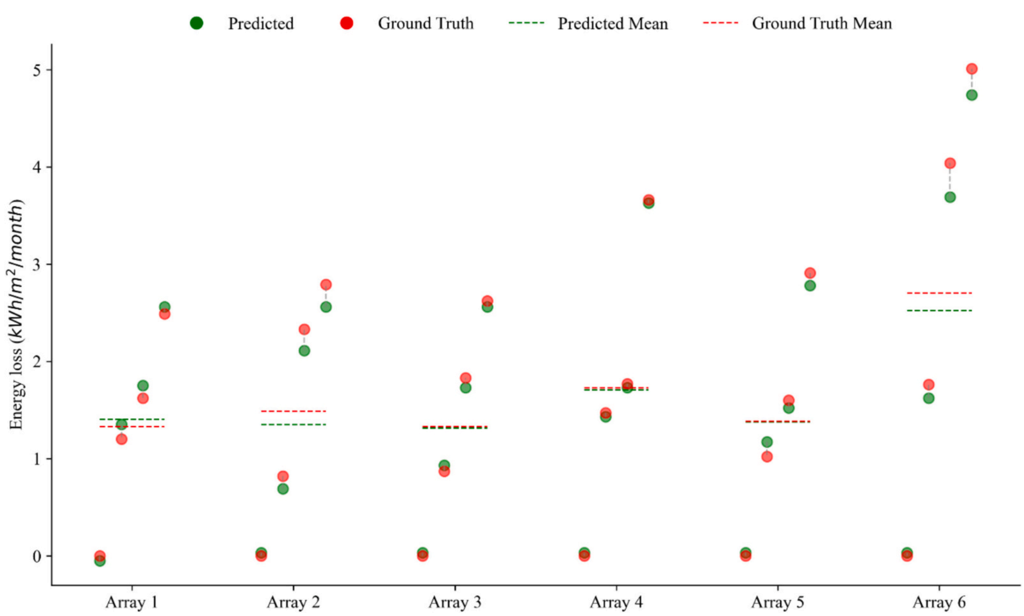

Fig. 13. Energy loss analysis due to snow for PV arrays.

ground-truth measurements and the U-Net model's predictions. The maximum energy losses per m^2 were reported for Array 6 due to its larger snow cover areas showing a predicted mean energy loss of 2.52 kWh/m^2/month and ground-truth loss of 2.70 kWh/m^2/month resulting in an error of 0.20 kWh/m^2/month. Array 1 and 2 showed mean energy losses of 1.48 and 1.50 kWh/m^2/month. Array 3 showed a mean energy loss of 1.46 kWh/m^2/month, while Array 4's analysis revealed mean energy losses of 1.57 kWh/m^2/month for predicted data and 0.38 kWh/m^2/month for ground-truth. This was attributed to Array 6 having the highest snow-coverage area percentage with a maximum value of 59%. The mean error across the arrays was calculated to be 0.07 kWh/m^2/month, highlighting the precision of the developed method in estimating energy losses due to snow coverage on solar panels.

The highest PV energy output was observed with snow removal conducted early in the morning, which allowed for maximum solar gain throughout the day. This scenario served as the baseline, where any delay in snow removal resulted in partial obstruction of solar energy due to snow accumulation on the PV panels. Fig. 14 illustrates the energy loss from PV arrays due to delayed snow removal across three cases. In Case 1, where snow removal occurred at 11 am, the panels were exposed to 6 h of solar radiation, from 11 am to 5 pm. In Case 2, snow removal at 1 pm allowed for 4 h of exposure, from 1 pm to 5 pm. Case 3, with snow removal at 3 pm, resulted in only 2 h of exposure, from 3 PM to 5 pm. These cases were applied to the examined six arrays, and energy production was compared to the baseline to estimate the energy loss attributed to snow coverage at different removal times. Array 6 experienced the highest energy loss, recording reductions of 68 Wh/m^2, 139 Wh/m^2, and 195 Wh/m^2 for the three cases, respectively. This showed that a one-hour delay in snow removal reduced PV production at an average rate of 32 Wh/m^2 per hour. The significant energy loss in Array 6 can be attributed to its maximum snow coverage of 59% as observed in Fig. 10. Conversely, Array 2 exhibited the minimum energy loss, with reductions of 21 Wh/m^2, 43 Wh/m^2, and 60 Wh/m^2 for the three cases, indicating an average reduction of 10 Wh/m^2 per hour for each hour of delay in snow removal. The mean energy loss across all arrays was calculated to be 40 Wh/m^2 when snow was cleared at 11 am, 90 Wh/m^2 when cleared at 1 pm, and 130 Wh/m^2 when cleared at 3 pm. Prompt snow removal is thus essential in the early morning to minimize energy losses and maximize the solar exposure of PV, thereby enhancing energy yield and overall performance.

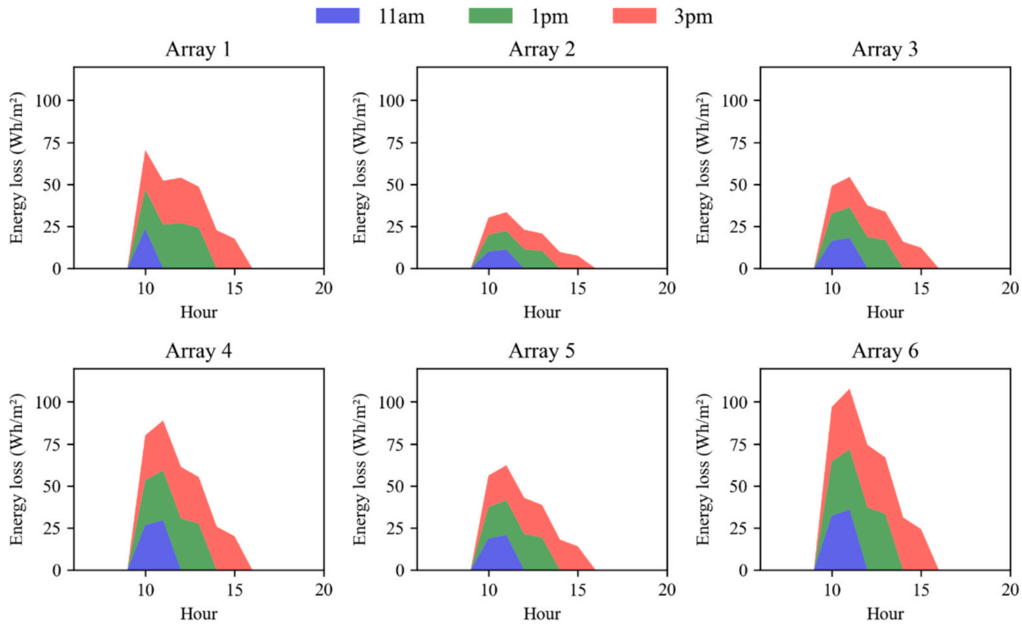

Fig. 14. Energy loss as a function of duration of snow cover.

Several key statistical metrics that highlight the accuracy and reliability of the predictions have been calculated based on the calculations as shown in Fig. 15. A sequence of images of PV panels with snow cover ranging from 0 to 100% were selected and compared with the results obtained in another study in a cold climate [79]. The Mean Absolute Error (MAE) was found to be 2.88%, indicating that, on average, the predicted snow coverage deviates from the actual snow cover. The Mean Squared Error (MSE) and the Root Mean Squared Error (RMSE) were calculated as 16.40% and 4.05%, respectively, with the RMSE providing insight into the standard deviation of the prediction errors. The model achieved an R-squared (R^2) value of 0.98, suggesting that 98% of the variance in the actual snow coverage is predictable from the developed model. Significant deviations were noted in the model's predictions for snow coverage below 20%. These metrics collectively support the model's high accuracy and its potential utility in applications related to monitoring and managing snow coverage on PV installations.

4. Conclusion

The increasing demand for sustainable and renewable energy sources has led to a significant increase in the use of solar PV systems in recent years. However, snow deposition in cold climates is one of the main factors that can reduce PV production, resulting in electricity generation losses of over 25% during the winter months. To address this problem, an automated detection system that utilizes deep learning and computer vision techniques is implemented. This research aimed to detect, and percentage of PV area covered by snow through the implementation of a deep learning-based automated detection system. A novel algorithm

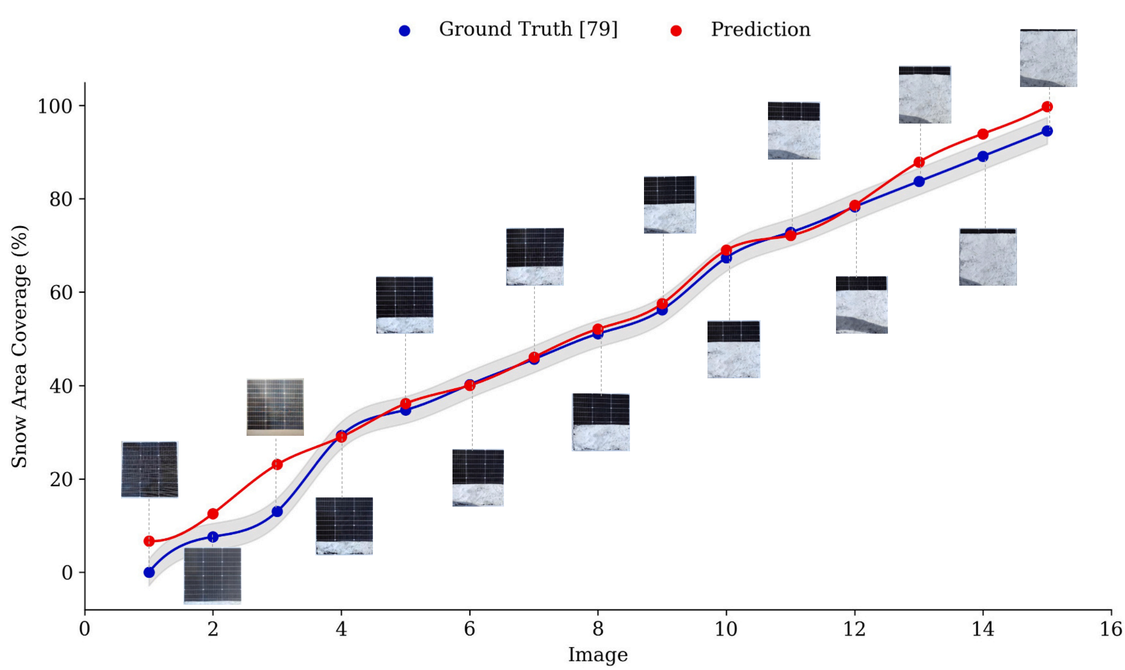

Fig. 15. Validation of snow area prediction based on trained model.

incorporating U-Net for image segmentation and timelapse imaging was proposed to accurately determine the snow-covered area of the solar panels. The method achieved a mean error of 2.88% when the results were compared to a set of 16 images captured from PV panels at different snow accumulation levels [79]. A comparative analysis was performed between major computer vision and image processing methods and deep learning models, revealing that deep learning models, with a Dice coefficient of 0.91, outperformed image processing methods, which achieved a Dice coefficient of 0.63, in image segmentation tasks. Energy losses were also estimated from the images with mean energy losses of 1.5 kWh/m^2/month from six different PV arrays. Moreover, the duration of snow cover was analyzed as a function of energy loss to understand the impact of different snow removal times on overall PV performance.

This study has noted limitations that could impact its broader applicability. The methodology accounts for the amount of snow coverage and the impact on string design losses, covering aspects such as module mismatch, diodes, DC wiring, soiling, and reflection losses (Incidence Angle Modifier, IAM); however, the location/pattern of snow coverage on the PV panels was simplified to consider one pattern per array with a total of six different patterns for six arrays. Another limitation lies in the requirement for a reference image of the PV panels without snow cover for each array. This prerequisite could potentially limit the system's utility in dynamic environments where pre-snowfall images are not readily available or cannot be captured under consistent conditions. Additionally, the performance and accuracy of the model could be influenced by various external factors such as lighting conditions, shadows, and the angle of image capture, which may not always be controllable or consistent across different PV installations.

Future research could focus on overcoming these limitations, potentially through the development of a more adaptive model that can perform accurately without the need for reference images and under varied environmental conditions. It is also recommended for future research to develop an autonomous system for cleaning PV panels and ensuring their efficiency during snow accumulation using the trained deep-learning model. The model used to calculate the amount of energy losses is a simplified version where a direct relation between A_{snow} and E_{loss} is assumed. However, in real-world scenarios, the relationship is not necessarily linear, as partial shading will decrease the production of the whole cell of the Array. Therefore, a more robust method of calculating E_{loss} should be developed for future studies.

CRediT authorship contribution statement

Mohamad T. Araji: Writing – original draft, Visualization, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Ali Waqas**: Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis. **Rahmat Ali**: Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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